Srikar Arani

Applied Space Research and Embedded Software Engineer, Johns Hopkins University Applied Physics Lab
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RESEARCH INTERESTS

Modeling, analysis, and design of verifiable machine learning models and autonomous systems. Topics of interest include low-SWaP edge computing, model verification and explainability, resilient cyber-physical systems, scientific computing, as well as the design of deep-learning accelerators and dataflows.

EDUCATION

University of California, Irvine 2024
Master of Science in Computer Engineering Irvine, CA
Thesis Title: Neuroflow: Deciphering Boiling Patterns From Neuromorphic Events
Advisors: Aparna Chandramowlishwaran, PhD and Yoonjin Won, PhD

University of California, Santa Barbara 2022
Bachelor of Science: Computer Engineering Santa Barbara, CA
Capstone Project: Diagnostic Apparatus for Fibrinogen Detection in Trauma Patients

ACADEMIC EXPERIENCE

Graduate Research Assistant Irvine, CA
University of California, Irvine September 2022 — March 2024

- Developed a real-time framework based on signals from neuromorphic sensors for flow regime classification under Professors Won and Chandramowlishwaran
- Compared performance of traditional high frame-rate cameras and optical convolutional neural netwroks against event-based approaches, including event-based versions of CNNs, SNNs, LSTMs and a novel Fourier K-Nearest Neighbors approach
- Final event-based LSTM approach achieved 97.6% accuracy with just 0.28 ms of processing time, running directly on raw event data without requiring preprocessing or specialized hardware.

PROFESSIONAL EXPERIENCE

Johns Hopkins University Applied Physics Lab Laurel, MD
Applied Space Research and Embedded DSP Software Engineer July 2024 — Present

- Designed Software-in-the-Loop environment for processing Air Force simulation data and generating track information for distributed Space Force constellation
- Interfaced directly with program management and sponsors for high-level system design problems, with particular focus to bottlenecks in software architecture and networking efficiency
- Lead developer for semi-analytic high energy-density, hyper-velocity impact physics modeling software
- Lead flight software engineer for multiple projects under different government sponsors, responsible for connecting GNC, PNT, and instrument software together for space research missions
- Lead developer for embedded linux system for US Government sponsored tracking effort

Internal Research and Development Grants

- Built machine learning model to issue real-time adjustments to ultra-stable oscillators, modeling clock hysteresis as a function of environmental effects
- Designed framework for Spiking Neural Networks to be used on space research missions for processing of event camera data
- Created software library for internal APL event-camera processing on space flight software
- Designed and implemented embedded software for low SwaP All-Band RF Transceiver

Appfolio Inc. Goleta, CA
Software Quality Assurance Engineer June 2019 — September 2022

- Designed and tested in an agile environment and helped push hundreds of projects and stories to users
- Worked on product security, while also refactoring hundreds of thousands of lines of tech debt to improve system software architecture

PUBLICATIONS

Journal paper
Published

- Chang S, Arani S, Nuthalapati NS, et al. EventFlow: Real-time neuromorphic event-driven classification of two-phase boiling flow regimes. Droplet. 2026; 5:e70066. https://doi.org/10.1002/dro2.70066

PROFESSIONAL HONORS

- **JHU/APL Space Sector Special Acheivement Award** for rapid and accurate design of modeling and simulation tool for National Security Space Missions (2023)

PROJECTS

Diagnostic Apparatus for Fibrinogen Detection in Trauma Patients 2021 — 2022
UCSB Capstone Project, Computer Engineering

- Designed and implemented a diagnostic tool for the detection of fibrinogen biomarker in trauma patients for early coagulopathy detection
- Personally responsible for working with and explaining technical details to sponsor
- Designed PCB and sensor protocols, making use of SPI, I2C, and UART for various sensors in apparatus
- Developed working prototype that earned third place in UC Santa Barbara's Capstone Presentation

MIPS Dual-Issue Processor 2021

- Designed and developed Dual-Issue Superscalar MIPS Pipelined processor capable of handling R, I and J type instructions
- Designed all components with verilog, capable of reading and processing machine code
- Featured two-level branch-prediction and a working cache

TECHNICAL SKILLS

- **Development:** C/C++, Python, Verilog HDL, MATLAB, Xilinx Vivado/Petalinux, Bash, SPICE
- **Frameworks:** PyTorch, TensorFlow, Z3 Theorem Prover
- **Miscellaneous:** Git, DevOps, Network Programming, Slurm